

EXPERIMENTAL PREDICTIONS OF THE SNYDERITE COHERENCE-IGNITION FRAMEWORK

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The Snyderite framework introduces two quantities absent from classical plasma physics: a **coherence field $C(t)$** governing the organizational integrity of the plasma, and a **surprise term $S = (1-C) \times \eta$** acting as an information-theoretic damper on energy release. Together they yield a modified ignition criterion and three categories of falsifiable experimental prediction. Each prediction is stated in a form directly testable on existing or near-term fusion experimental platforms including JET, ITER, NIF, and TAE Technologies.

I. THE MODIFIED LAWSON CRITERION (Snyderite Ignition Condition)

Classical Lawson criterion for D-T fusion requires: $n T \tau_E > 3 \times 10^{21} \text{ keV s m}^{-3}$. The Snyderite framework adds a necessary second condition on coherence:

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IGNITION iff { n T tau_E > Lawson_threshold } AND { C > C_min }
where C_min ~ 0.90 (minimum organizational coherence for sustained burn)
drive = eta_i * C * [M > M_min] --> drive = 0 when C < C_min OR M < M_min
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The physical interpretation: **Lawson governs whether sufficient thermal energy exists. Snyderite governs whether the plasma is organized enough to burn it.** Below $C_{\min}$, turbulent scattering dissipates energy faster than fusion reactions can sustain — the plasma has fuel but lacks the structure to use it. This is not a new physical force. It is a recognition that anomalous transport (the dominant loss channel in all current tokamaks) is an information-theoretic phenomenon: disorder bleeding energy from organized modes into thermal noise.

II. EXPERIMENTAL PREDICTION 1 — COHERENCE THRESHOLD FOR IGNITION

Statement: Plasmas that satisfy Lawson but fail to ignite will be measurably less coherent (lower C) than plasmas at comparable Lawson parameters that do sustain burn. C is operationally defined as the normalized cross-correlation amplitude of turbulent fluctuations across spatial scales.

Measurement	Cross-correlation $C(x,x')$ of density/temperature fluctuations
Platform	JET / ITER (existing reflectometry + BES diagnostic suite)
Test condition	Compare plasmas: same n, T, τ_E — different confinement quality
Prediction	Igniting plasmas will show $C > 0.90$ measured coherence
Null result	If C is uncorrelated with ignition at fixed Lawson: framework fails
Signature	C will show a threshold transition, not a smooth gradient

III. EXPERIMENTAL PREDICTION 2 — COHERENCE-WEIGHTED YIELD

Statement: For two plasmas with identical Lawson triple products, the plasma with higher measured coherence C will produce higher fusion yield. The yield ratio scales as $C_{\text{high}} / C_{\text{low}}$, not as the ratio of any thermal parameter.

C value	Surprise S	Drive factor	Yield loss	Physical meaning
0.90	$0.050 \times \eta$	$0.90 \times \text{drive}$	10%	Strong turbulent scattering
0.95	$0.025 \times \eta$	$0.95 \times \text{drive}$	5%	Moderate anomalous transport
0.99	$0.005 \times \eta$	$0.99 \times \text{drive}$	1%	Weak residual turbulence
0.999	$0.0005 \times \eta$	$0.999 \times \text{drive}$	0.1%	Near-laminar organized burn

Platform	NIF (inertial confinement) or SPARC (high-field tokamak)
Measurement	Yield per shot vs. measured mode coherence (Thomson scattering)
Prediction	$dY/dC > 0$ at fixed n, T, τ_E — yield rises with coherence
Null result	If yield is uncorrelated with C at fixed Lawson: framework fails

IV. EXPERIMENTAL PREDICTION 3 — SELF-REGULATING TRANSPORT SUPPRESSION

Statement: A plasma governed by SMEM dynamics will exhibit self-regulating anomalous transport. When turbulence degrades C, the surprise term $S = (1-C) \times \eta$ rises, reducing the effective drive, which reduces turbulence, allowing C to recover. This produces a **homeostatic oscillation** in transport coefficients — not a monotonic decay — observable as quasi-periodic suppression and recovery cycles without external intervention.

Measurement	Turbulent transport coefficient χ_i vs. time at fixed heating power
Platform	DIII-D / ASDEX-U (rich diagnostic suite, variable NBI power)
Prediction	χ_i will show quasi-periodic suppression cycles, period $\sim \tau_C$
Prediction	Suppression depth will correlate with $(1-C_{\text{measured}})$, not with T or n
Prediction	Increasing heating drive (η_i) will shorten cycle period, not deepen suppression
Null result	If χ_i is monotonic or uncorrelated with coherence cycles: fails
Distinguisher	Classical ITG predicts χ_i proportional to η_i . SMEM predicts saturation.

V. THE SNYDERITE IGNITION MAP — 2D PARAMETER SPACE

Classical ignition maps plot $n\tau_E$ vs. T (Lawson space). The Snyderite framework adds a third axis: C. This produces a 3D ignition volume with the following structure:

Region A	Lawson satisfied, $C > C_{\text{min}}$	--> IGNITION — sustained burn
Region B	Lawson satisfied, $C < C_{\text{min}}$	--> FAILED IGNITION — Lawson-blind experiments land here
Region C	Lawson unsatisfied, $C > C_{\text{min}}$	--> SUB-IGNITION — coherent but insufficient energy
Region D	Lawson unsatisfied, $C < C_{\text{min}}$	--> COLD PLASMA — no ignition by any path

Region B is the critical prediction. SMEM predicts that a non-trivial fraction of historical failed ignition attempts in tokamaks and laser-fusion experiments occupied Region B: Lawson was met or nearly met, but C was below threshold. This would explain anomalous transport losses that classical models attribute to unresolved physics. The Snyderite framework names them: they are coherence deficits.

VI. ANEUTRONIC FUSION (p-11B) — SPECIAL CASE

The p-¹¹B reaction (8.7 MeV, no neutrons) has the most to gain from the Snyderite framework. Classical ignition for p-¹¹B requires plasma temperatures of 200-300 keV — far beyond current tokamak capability. However, the coherence-gated drive predicts that **a sufficiently organized p-¹¹B plasma at lower temperature may sustain partial burn** through coherence amplification of the reaction rate. This is testable at TAE Technologies (Norman, CA) using their field-reversed configuration platform. Prediction: p-¹¹B yield will show stronger coherence dependence than D-T yield, because the lower thermal drive makes the C-weighting term proportionally more significant.

VII. SUMMARY OF FALSIFIABLE PREDICTIONS

P1	Ignition threshold $C > C_{\text{min}}$ required independent of Lawson — JET/ITER
P2	Yield-coherence scaling $\eta/dC > 0$ at fixed n,T, τ_E — NIF/SPARC
P3	Transport self-regulation χ_i shows quasi-periodic suppression cycles — DIII-D
P4	Region B identification: Historical failed ignitions reanalyzed for C deficit
P5	p-11B coherence amplification: Stronger C-dependence in aneutronic than D-T yield — TAE

Each prediction above is falsifiable by existing experimental apparatus. None requires new physics beyond the identification of plasma coherence C as a dynamical variable with its own equation of motion. The Snyderite framework does not replace classical plasma physics. It adds one variable — C — and one equation — $dC/dt = f(C, S, \Phi_K)$ — and derives from that addition a set of consequences that classical theory cannot predict. If any prediction fails under controlled conditions, the framework is falsified in that domain and must be revised. **That is the agreement.**

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